04-31 第1篇

阅读理解（说明文）

① Have you ever noticed that a piece of iron (铁) left outside in the rain gradually turns reddish-brown? This reddish-brown stuff is called rust (铁锈). Rusting is a very common chemical reaction (化学反应) in our daily lives. But how exactly does it happen, and why is it a problem?

② For iron to rust, three things must be present at the same time: iron, oxygen (氧气), and water. When an iron object, like a bicycle chain or a nail, is exposed (暴露) to the air and moisture (水分), the oxygen in the air combines with the iron. The water acts as a helper to speed up this process. Together, they create a new substance—iron oxide (氧化铁), which we commonly call rust.

③ Once rust starts to form, it doesn't stop easily. Unlike some metals, such as aluminum (铝), which form a protective layer when they react with oxygen, rust is soft and crumbles (碎裂) easily. As the rust falls off, new iron is exposed to the air and water, and the rusting process continues. Over time, a thick piece of strong iron can be completely eaten away and turn into a pile of red powder (粉末).

④ Rusting is a huge problem because it destroys (破坏) metal objects. Bridges, cars, ships, and pipes are often made of iron or steel (which contains a lot of iron). If they rust, they become weak and dangerous. Every year, people spend billions of dollars fixing or replacing rusted machines and buildings.

⑤ To fight against rust, people have invented several methods. The most common way is to paint the iron. A layer of paint creates a wall between the iron and the air, stopping oxygen and water from reaching the metal. Another method is to cover the iron with a layer of zinc (锌), a process called galvanization (镀锌). By understanding how rust works, we can better protect our important metal structures.

注释： rust 铁锈，生锈；reaction 反应；expose 暴露；moisture 水分；crumble 碎裂；powder 粉末

选择题

1. What three things are needed for iron to rust according to paragraph ②? A. Iron, carbon dioxide, and hot air. B. Iron, sunlight, and strong wind. C. Iron, oxygen, and water. D. Iron, zinc, and a piece of cloth.

2. How does water act during the rusting process? A. It stops oxygen from reaching the iron. B. It acts as a helper to speed up the process. C. It washes the red rust

away completely. D. It turns the iron into hard aluminum.

3. Why does iron continue to rust until it is completely eaten away? A. Because rust is a hard protective layer that covers the iron. B. Because iron loves to eat red powder. C. Because rust is soft and crumbles, exposing new iron to the air and water. D. Because aluminum keeps pushing oxygen into the iron.

4. Form paragraph ④, why is rusting considered a huge problem in our daily lives? A. Because it makes the air dirty and hard to breathe. B. Because it turns cheap iron into expensive gold. C. Because it destroys strong metal objects like bridges and cars, making them weak and dangerous. D. Because it forces people to build wooden bridges instead.

5. How does painting iron help prevent (防止) it from rusting? A. The paint creates a wall that stops oxygen and water from reaching the metal. B. The smell of the paint scares the water away. C. The paint absorbs completely into the iron and makes it harder. D. The paint changes the iron into zinc.

详细解析

1. 【答案】C

【题型】 细节理解

【定位】 第②段第1句 "For iron to rust, three things must be present at the same time: iron, oxygen (氧气), and water."

【解析】 原文中明确列出了铁生锈必须同时具备的三个条件：铁、氧气（oxygen）和水（water）。A中二氧化碳、B中强风阳光、D中锌布都不正确。故选C。

2. 【答案】B

【题型】 细节理解

【定位】 第②段第3句 "The water acts as a helper to speed up this process."

【解析】 根据原文描述，水分在生锈过程中的作用是充当一个“帮手（helper）”来加速（speed up）这一过程。这与B项一致。A项阻断氧气那是油漆的作用，C项洗掉铁锈、D项变成铝均是无中生有。故选B。

3. 【答案】C

【题型】 细节理解推断

【定位】 第③段 "...rust is soft and crumbles (碎裂) easily. As the rust falls off, new iron is exposed to the air and water, and the rusting process continues."

【解析】 本段解释铁会一直生锈直到被完全毁掉的原因：与铝不同，铁锈松软易碎（crumbles easily），脱落后新的铁会再次暴露在空气和水中，使得这一过程周而复始。因此C项完美呈现了这层逻辑。A项说形成坚硬保护层，这恰好是张冠李戴（描写铝的）。故选C。

4. 【答案】C

【题型】 细节理解

【定位】 第④段第1-3句 "Rusting is a huge problem because it destroys (破坏) metal objects. Bridges, cars, ships, and pipes... If they rust, they become weak and dangerous."

【解析】 生锈之所以是一个巨大的问题，是因为它会破坏桥梁、汽车等金属物件，使它们变得脆弱和危险。选项C（因为它破坏了像桥梁和汽车这样坚固的金属物体，使它们变得脆弱危险）与原文意思相符。故选C。

5. 【答案】A

【题型】 细节理解

【定位】 第⑤段第2-3句 "The most common way is to paint the iron. A layer of paint creates a wall between the iron and the air, stopping oxygen and water from reaching the metal."

【解析】 第五段提及防锈方法时指出，给铁涂漆可以通过在铁与空气之间建起一道墙，从而阻止氧气和水分接触到金属。这直接对应A选项的意思。闻味驱水（B）、吸入变硬（C）、变成锌（D，镀锌是另外的方法而不是油漆的作用）皆滑稽荒谬。故选A。